

Unit 10, Lesson 10: Using Tables to Determine if a Function is Linear

Benchmark

MA.8.F.1.2 Given a function defined by a graph or an equation, determine whether the function is a linear function. Given an input-output table, determine whether it could represent a linear function.

Learning Targets

- ☐ I can determine whether a function could represent a linear function given a table of values.
- ☐ I can determine whether a function is a linear function given an equation or graph.

10.10.1 Warm-Up: Which One Doesn't Belong?

Four number patterns are shown.

Pattern A	Pattern B
5, 9, 13, 17, 21, ...	5, 15, 45, 135, 405, ...
Pattern C	Pattern D
5, 2, -1, -4, -7, ...	10, 13, 16, 19, 22, ...

1. Complete the statement to explain which pattern doesn't belong.

☐ Pattern A

☐ Pattern B

☐ Pattern C

☐ Pattern D

doesn't belong because _____

_____.

2. Find a classmate who selected a different pattern that doesn't belong. Write their explanation in the table. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary of their explanation stating that your summary is correct.

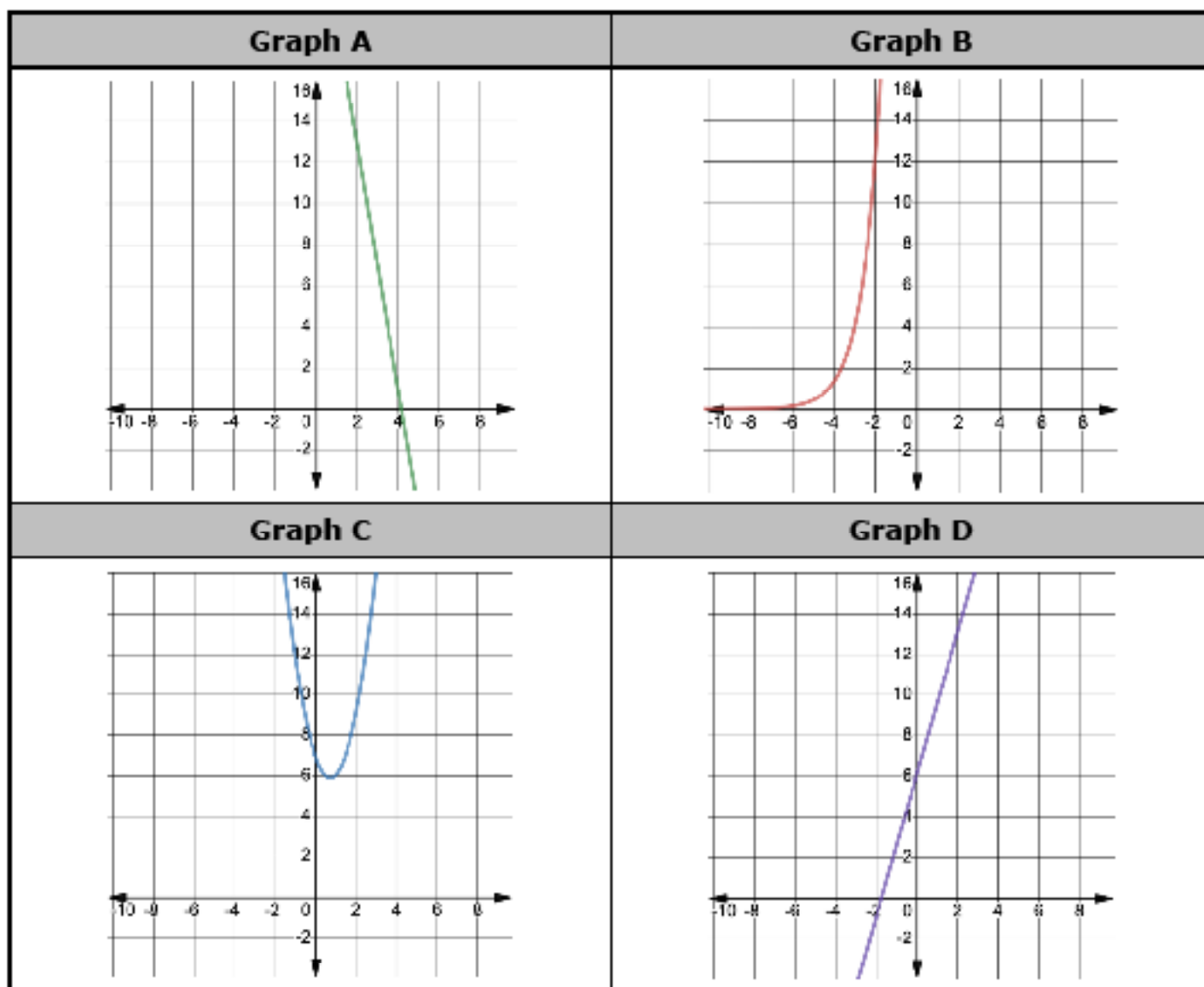
My summary of _____'s description	Initials

10.10.2 Guided Practice: Exploring Tables to Determine Linear Functions

1. Complete the description observed in the table of values for each function.

Function		Description												
A	<table><tr><th>x</th><th>y</th></tr><tr><td>1</td><td>19</td></tr><tr><td>2</td><td>13</td></tr><tr><td>3</td><td>7</td></tr><tr><td>4</td><td>1</td></tr><tr><td>5</td><td>-5</td></tr></table>	x	y	1	19	2	13	3	7	4	1	5	-5	The rate of change, $\frac{\Delta y}{\Delta x}$, ☐ is ☐ is not constant between each pair of values.
x	y													
1	19													
2	13													
3	7													
4	1													
5	-5													
B	<table><tr><th>x</th><th>y</th></tr><tr><td>-3</td><td>4</td></tr><tr><td>-2</td><td>12</td></tr><tr><td>-1</td><td>36</td></tr><tr><td>0</td><td>108</td></tr><tr><td>1</td><td>324</td></tr></table>	x	y	-3	4	-2	12	-1	36	0	108	1	324	The rate of change, $\frac{\Delta y}{\Delta x}$, ☐ is ☐ is not constant between each pair of values.
x	y													
-3	4													
-2	12													
-1	36													
0	108													
1	324													
C	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>21</td></tr><tr><td>-1</td><td>12</td></tr><tr><td>0</td><td>7</td></tr><tr><td>1</td><td>6</td></tr><tr><td>2</td><td>9</td></tr></table>	x	y	-2	21	-1	12	0	7	1	6	2	9	The rate of change, $\frac{\Delta y}{\Delta x}$, ☐ is ☐ is not constant between each pair of values.
x	y													
-2	21													
-1	12													
0	7													
1	6													
2	9													
D	<table><tr><th>x</th><th>y</th></tr><tr><td>-4</td><td>-8</td></tr><tr><td>-2</td><td>-1</td></tr><tr><td>0</td><td>6</td></tr><tr><td>2</td><td>13</td></tr><tr><td>4</td><td>20</td></tr></table>	x	y	-4	-8	-2	-1	0	6	2	13	4	20	The rate of change, $\frac{\Delta y}{\Delta x}$, ☐ is ☐ is not constant between each pair of values.
x	y													
-4	-8													
-2	-1													
0	6													
2	13													
4	20													

2. For each table that has a constant rate of change, complete the statement.
- a. In table _____ as the x -values increase by _____, the y -values _____.
- b. In table _____ as the x -values increase by _____, the y -values _____.
3. The graphs for each of the functions represented by tables are shown. Put an asterisk (*) next the graphs that are linear functions.



10.10.3 Guided Instruction: Identifying Linear Functions from Tables

1. Notice that for the linear functions, as the x -values in the tables increase by the same amount, the y -values either increase or decrease by

_____.

2. In each example of a linear function, the rate of change, $\frac{\Delta y}{\Delta x}$, is

_____.



A **linear function** is a function that has a constant rate of change.

3. Determine which tables of values show a rate of change, $\frac{\Delta y}{\Delta x}$, that is constant.

Table E	Table F	Table G																														
<table><tr><th>x</th><th>y</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>4</td></tr><tr><td>2</td><td>7</td></tr><tr><td>3</td><td>10</td></tr></table>	x	y	0	1	1	4	2	7	3	10	<table><tr><th>x</th><th>y</th></tr><tr><td>-3</td><td>3</td></tr><tr><td>0</td><td>1</td></tr><tr><td>3</td><td>-1</td></tr><tr><td>9</td><td>-5</td></tr></table>	x	y	-3	3	0	1	3	-1	9	-5	<table><tr><th>x</th><th>y</th></tr><tr><td>2</td><td>-4</td></tr><tr><td>4</td><td>0</td></tr><tr><td>6</td><td>4</td></tr><tr><td>8</td><td>8</td></tr></table>	x	y	2	-4	4	0	6	4	8	8
x	y																															
0	1																															
1	4																															
2	7																															
3	10																															
x	y																															
-3	3																															
0	1																															
3	-1																															
9	-5																															
x	y																															
2	-4																															
4	0																															
6	4																															
8	8																															

4. The graphs for each of the functions E, F, and G are shown with their corresponding table of values. If the function represents a linear function, circle the function's letter.

Function	Table	Graph										
E	<table><tr><th>x</th><th>y</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>4</td></tr><tr><td>2</td><td>7</td></tr><tr><td>3</td><td>10</td></tr></table>	x	y	0	1	1	4	2	7	3	10	
x	y											
0	1											
1	4											
2	7											
3	10											
F	<table><tr><th>x</th><th>y</th></tr><tr><td>-3</td><td>3</td></tr><tr><td>0</td><td>1</td></tr><tr><td>3</td><td>-1</td></tr><tr><td>9</td><td>-5</td></tr></table>	x	y	-3	3	0	1	3	-1	9	-5	
x	y											
-3	3											
0	1											
3	-1											
9	-5											
G	<table><tr><th>x</th><th>y</th></tr><tr><td>2</td><td>-4</td></tr><tr><td>4</td><td>0</td></tr><tr><td>6</td><td>4</td></tr><tr><td>8</td><td>8</td></tr></table>	x	y	2	-4	4	0	6	4	8	8	
x	y											
2	-4											
4	0											
6	4											
8	8											

- 5. In Function E, describe what is different in the graph that was not shown in the table.**

- 6. In Function G, describe what is different in the graph that was not shown in the table.**

A table of values only gives some of the points on the graph of a function. This is not always enough information to conclude that a function is linear. Therefore, from a table of values, it can only be concluded that a function *could* represent a linear function.

10.10.4 Your Turn!

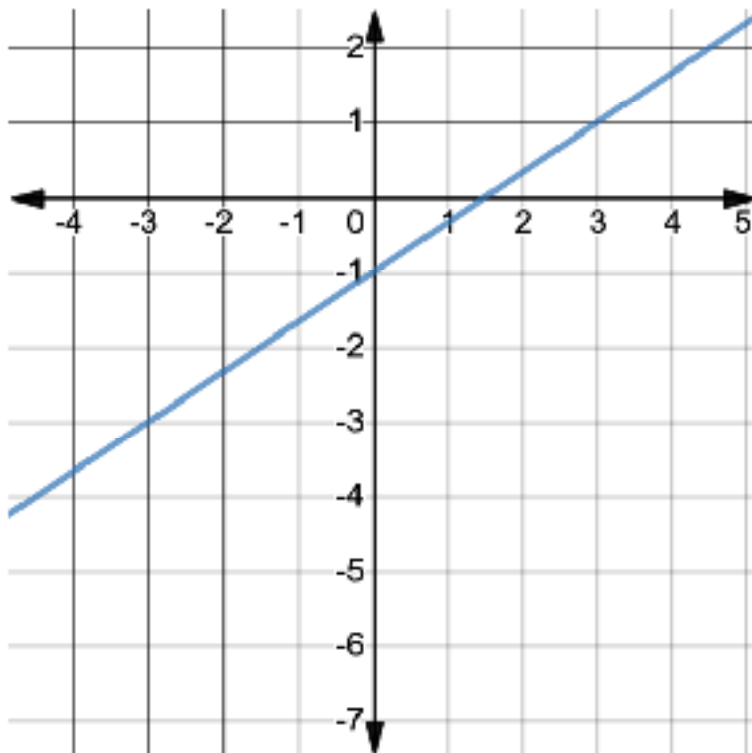
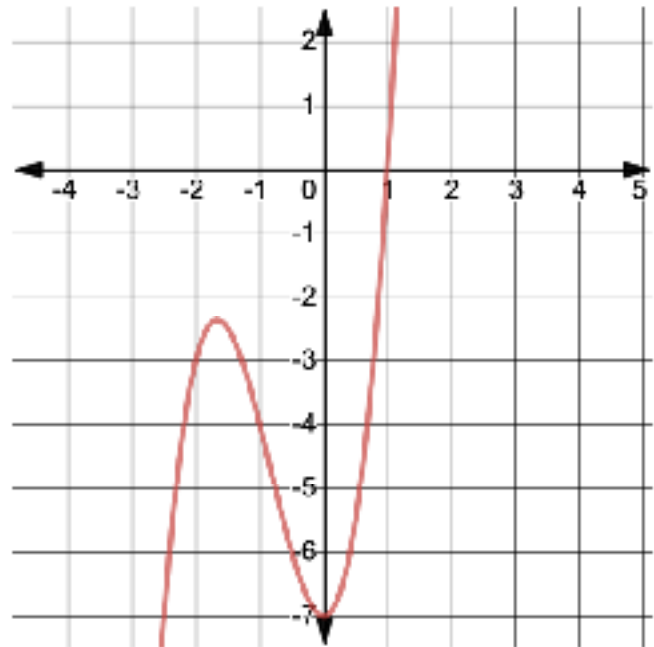
1. Determine which relations could be linear functions. Show your work.

Relation	Description										
<table><tr><th><i>x</i></th><th><i>y</i></th></tr><tr><td>-7</td><td>4</td></tr><tr><td>-3</td><td>3</td></tr><tr><td>1</td><td>2</td></tr><tr><td>7</td><td>1</td></tr></table>	<i>x</i>	<i>y</i>	-7	4	-3	3	1	2	7	1	The relation ☐ could be ☐ cannot be a linear function because ...
<i>x</i>	<i>y</i>										
-7	4										
-3	3										
1	2										
7	1										
<table><tr><th><i>x</i></th><th><i>y</i></th></tr><tr><td>12</td><td>-2</td></tr><tr><td>8</td><td>1</td></tr><tr><td>4</td><td>4</td></tr><tr><td>0</td><td>7</td></tr></table>	<i>x</i>	<i>y</i>	12	-2	8	1	4	4	0	7	The relation ☐ could be ☐ cannot be a linear function because ...
<i>x</i>	<i>y</i>										
12	-2										
8	1										
4	4										
0	7										

2. Six different relations are shown. Circle the relations that represent linear functions.

$$y = 2.5x$$

$$y = \frac{8}{x} + 12$$

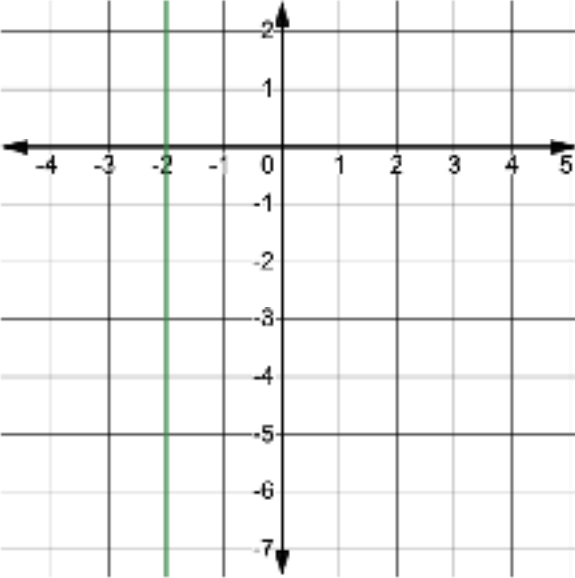


$$y = -4x^3 + 7$$

$$4.2x = 0.1 - 3y$$

10.10.5 Wrap-Up: Ticket Out the Door

1. Complete the table to describe each relation.

Relation	Description										
<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>-2</td></tr><tr><td>0</td><td>-4</td></tr><tr><td>2</td><td>-8</td></tr><tr><td>4</td><td>-16</td></tr></table>	x	y	-2	-2	0	-4	2	-8	4	-16	The relation ☐ could be ☐ cannot be a linear function.
x	y										
-2	-2										
0	-4										
2	-8										
4	-16										
<table><tr><th>x</th><th>y</th></tr><tr><td>-5</td><td>6</td></tr><tr><td>-2</td><td>2</td></tr><tr><td>1</td><td>-2</td></tr><tr><td>4</td><td>-6</td></tr></table>	x	y	-5	6	-2	2	1	-2	4	-6	The relation ☐ could be ☐ cannot be a linear function.
x	y										
-5	6										
-2	2										
1	-2										
4	-6										
$y = 7.2$	The relation ☐ is ☐ cannot be a linear function.										
	The relation ☐ is ☐ cannot be a linear function.										